

## Applied Statistics – Final Exam – 02 September 2022 – Spotify Top 50 Songs 2021

Info are available on the Top 50 most listened songs in the world for year 2021 in Spotify.

Please use your knowledge on multiple regression to answer to the following questions.

- 1) Can we predict the popularity of a song given the values of the other variables available in the dataset?
- 2) Given the other variables, is danceability significantly related to popularity?
- 3) If you wish to author a popular song on Spotify, which features would you mostly consider?

### *Data description*

Popularity - The higher the value the more popular the song is

Danceability - The higher the value, the easier it is to dance to this song

Energy - The energy of a song - the higher the value, the more energetic is the song

Key - the key the track is in. Integers map to pitches using standard Pitch Class notation. E.g. 0 = C, 1 = C#/D♭, 2 = D, and so on. If no key was detected, the value is -1 (range: -1; 11)

Loudness (dB) - The higher the value, the louder the song

Mode - indicates the modality (major or minor) of a track, the type of scale from which its melodic content is derived. Major is represented by 1 and minor is 0

Speechiness - The higher the value the more spoken word the song contains

Acousticness - The higher the value the more acoustic the song is

Instrumentalness - the number of vocals in a song. The closer the value to 1.0, the more instrumental the song is

Liveness - The higher the value, the more likely the song is a live recording

Valence - The higher the value, the more positive mood for the song

Tempo - the overall estimated tempo of a track in beats per minute (BPM)

Duration - duration of the song in ms

Time signature - an estimated time signature. The time signature (meter) is a notational convention to specify how many beats are in each bar (or measure). The time signature ranges from 3 to 7 indicating time signatures of "3/4", to "7/4"

**Provide a report of maximum 4 pages**

**The report must be uploaded on Google Classroom as a pdf file**